

Vector Calculus

Aayam Shrestha

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Introduction

These notes are composed from textbooks **Erwin Kreyszig, Advanced Engineering Mathematics**.

1 Vectors in 2-space and 3-space

3-space is three dimensional space and 2-space is the plane(special case of 3-space)

scalars are numbers with magnitude only and **vectors** are arrows having specified length(magnitude) pointing in certain direction.

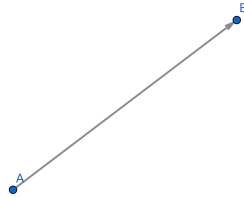


Figure 1: Vector

here, **A** is called the initial point(tail) and **B** is called the terminal point(tip).

Vectors are used in translation(displacement without rotation).
Two vectors $\vec{a}$ and $\vec{b}$ are said to be **equal**, if they have same length and direction.

Magnitude of $\vec{a}$ is written as $|\vec{a}|$ (**Euclidean Norm**). Unit vector has norm of 1.

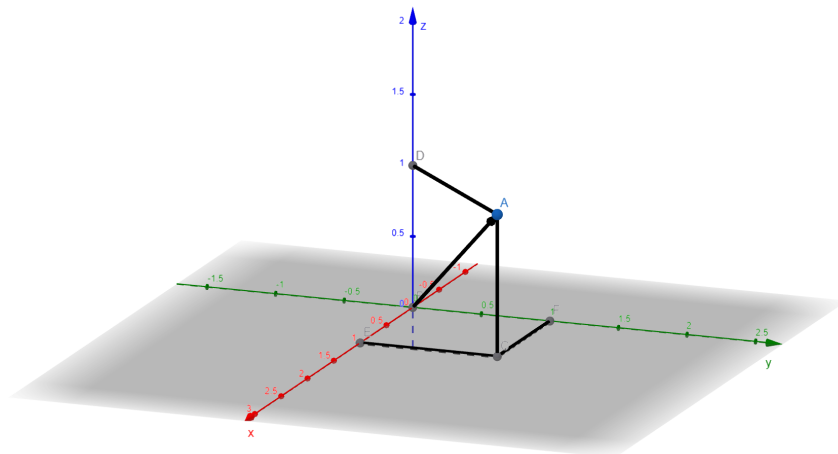


Figure 2: 3D vector

This figure shows the **position vector** $\vec{r} = (1, 1, 1)$. Position vector means vector with the tail as origin and tip as some point in $\mathbb{R}^3$.

Imagine two points in 3-space, $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ then components of $\vec{PQ}$ are $a_1 = x_2 - x_1, a_2 = y_2 - y_1, a_3 = z_2 - z_1$. $\vec{PQ} = \vec{a} = [a_1, a_2, a_3]$. $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ (**3D Pythagorus Theorem**)

1.1 Vector Addition(Resultant Vector)

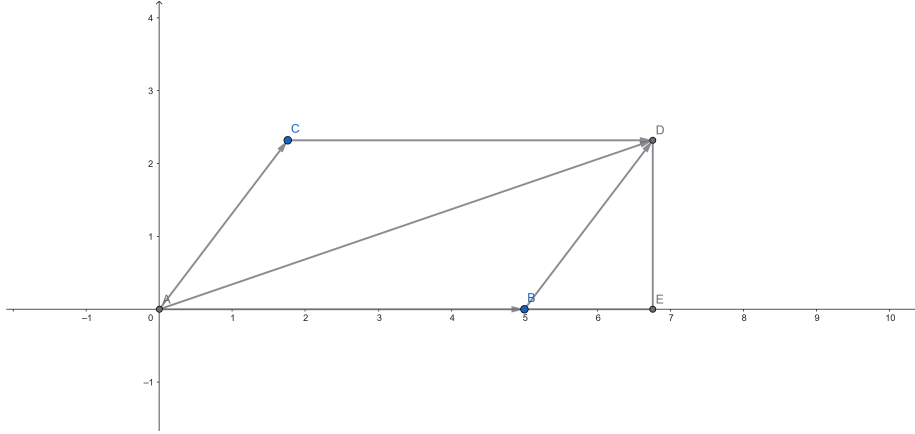


Figure 3: Resultant Vector

In this figure, we consider two vectors $\vec{AB}$ and $\vec{AC}$ with the angle between them being $\angle CAB = \theta$. Now, to find the **resultant** of these two vectors, we use **parallelogram law of vector addition**. This law basically says, if two vectors are thought of two adjacent sides of parallelogram, it's diagonal is the resultant. In this case diagonal ($\vec{AD}$) is resultant of $\vec{AB} + \vec{AC}$.

Now, let α be the angle made by resultant with the positive x-axis(horizontal in anticlockwise direction).

Drop $DE \perp$ x-axis, $\angle DBE = \angle CAB = \theta$ and $BD = AC$. Thus, $BE = BD \cos \theta$ and $DE = BD \sin \theta$.

$$\begin{aligned} AD^2 &= (AB + BE)^2 + DE^2 \\ &= (AB + BD \cos \theta)^2 + (BD \sin \theta)^2 \\ &= AB^2 + BD^2 \cos^2 \theta + BD^2 \sin^2 \theta + 2 \cdot AB \cdot BD \cdot \cos \theta \\ &= AB^2 + AC^2 + 2 \cdot AB \cdot AC \cdot \cos \theta \end{aligned}$$

$$\tan \alpha = \frac{b \sin \theta}{a + b \cos \theta}$$

Some Basic Properties of Vector Addition:

$$\begin{aligned} \vec{a} + \vec{b} &= \vec{b} + \vec{a} \\ (\vec{u} + \vec{v}) + \vec{w} &= \vec{u} + (\vec{v} + \vec{w}) \\ \vec{a} + \vec{0} &= \vec{0} + \vec{a} = \vec{a} \\ \vec{a} + (-\vec{a}) &= \vec{0} \end{aligned}$$

1.2 Scalar Multiplication (Multiplication by numbers)

$c\vec{a} = \{ca_1, ca_2, ca_3\}$, where $\vec{a} = \{a_1, a_2, a_3\}$

If $\vec{a} \neq \vec{0}$, then $c\vec{a}$ with $c > 0$ has direction of $\vec{a}$, and for $c < 0$, $c\vec{a}$ has opposite direction.

In all cases, $|c\vec{a}| = |c||\vec{a}|$ and $c\vec{a} = \vec{0}$ if $\vec{a} = \vec{0}$ or $c = 0$ (or both).

Some Basic Properties of Scalar Multiplication:

$$c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$$

$$(c + k)\vec{a} = c\vec{a} + k\vec{a}$$

$$c(k\vec{a}) = (ck)\vec{a}$$

$$1\vec{a} = \vec{a}$$

$$0\vec{a} = \vec{0}$$

$$(-1)\vec{a} = -\vec{a}$$

For Unit vectors $\vec{i}, \vec{j}$ and $\vec{k}, \vec{a} = \{a_1, a_2, a_3\} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$. $\vec{i} = [1, 0, 0], \vec{j} = [0, 1, 0]$ and $\vec{k} = [0, 0, 1]$. $\mathbb{R}^3$ has dimension 3. $\vec{i}, \vec{j}, \vec{k}$ is called a **standard basis** of $\mathbb{R}^3$.

Electrostatics: From Point Charges to Surface Integrals Physics and Mathematics Notes

2 The Principle of Superposition: Discrete Case

In electrostatics, the interaction between point charges is governed by Coulomb's Law. When multiple source charges $q_1, q_2, \dots, q_n$ are present, the total force and field at a test point are determined by the principle of superposition.

2.1 Total Electrostatic Force

The total force $\vec{F}$ exerted on a test charge q_0 located at position $\vec{r}$ by a system of n discrete charges q_i at positions $\vec{r}_i$ is the vector sum:

$$\vec{F}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_0 q_i}{|\vec{r} - \vec{r}_i|^2} \hat{u}_i$$

Where $\hat{u}_i$ is the unit vector directed from q_i toward q_0 .

2.2 The Electric Field

The electric field $\vec{E}$ is defined as the force per unit charge. By dividing the total force by the test charge q_0 , we obtain the field expression for the discrete case:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\vec{r} - \vec{r}_i|^2} \hat{u}_i$$

3 Continuous Charge Distributions

In many macroscopic systems, charge is distributed so densely that it is more practical to treat it as a continuum rather than a set of discrete points.

3.1 Defining Charge Density

We define the **average charge density** $\bar{\rho}_{\Delta V}$ over a small volume ΔV as:

$$\bar{\rho}_{\Delta V} = \frac{\Delta Q}{\Delta V}$$

To find the density at a specific point (x, y, z) , we take the limit as the volume shrinks to zero:

$$\rho(x, y, z) = \lim_{\Delta V \rightarrow 0} \frac{\Delta Q}{\Delta V} = \frac{dQ}{dV}$$

3.2 Total Charge and Resultant Field

The total charge Q contained within a volume V is the volume integral of the density:

$$Q = \iiint_V \rho(x, y, z) dV$$

Consequently, the electric field produced by this volume distribution is given by:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \iiint_V \frac{\rho(\vec{r}') \hat{u}(\vec{r}')}{|\vec{r} - \vec{r}'|^2} dV'$$

4 Surface Integrals and Gauss's Law

Gauss's Law is a fundamental tool relating the flux of the electric field through a closed surface to the charge enclosed within that surface.

$$\Phi_E = \iint_S \vec{E} \cdot \hat{n} dS = \frac{q_{enc}}{\epsilon_0}$$

4.1 Calculating the Unit Normal Vector $\hat{n}$

For a surface defined explicitly by the function $z = f(x, y)$, the unit normal vector $\hat{n}$ is essential for calculating the surface integral. We begin by defining tangent vectors in the x and y directions:

- **Tangent to XZ Plane:** $\vec{u} = \left[\hat{i} + \left(\frac{\partial f}{\partial x} \right) \hat{k} \right] u_x$
- **Tangent to YZ Plane:** $\vec{v} = \left[\hat{j} + \left(\frac{\partial f}{\partial y} \right) \hat{k} \right] v_y$

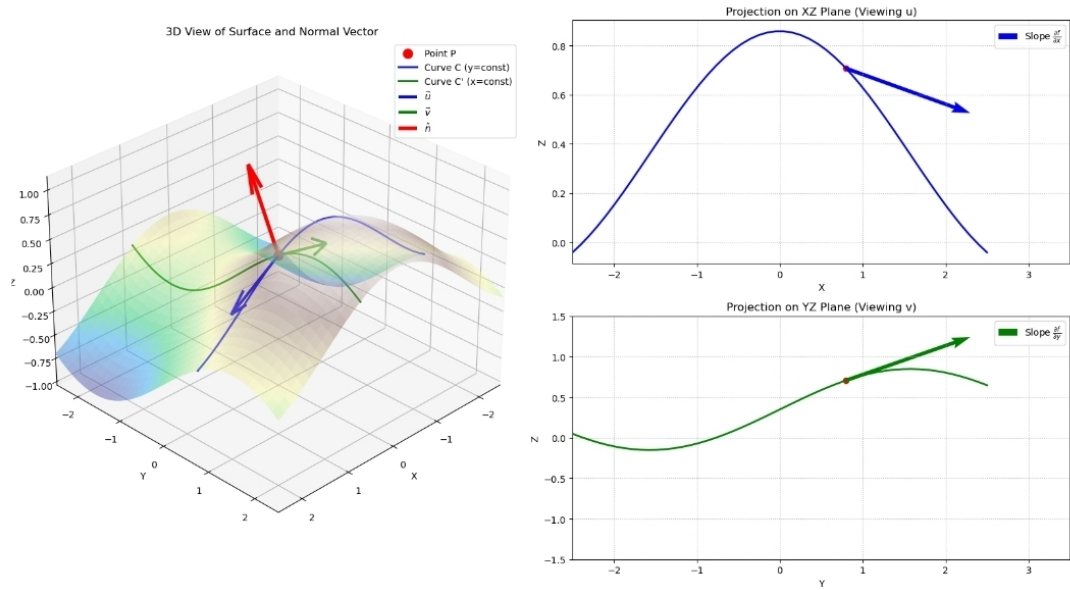


Figure 4: Normal Vector

The normal vector $\vec{N}$ is found via the cross product $\vec{u} \times \vec{v}$:

$$\vec{N} = \vec{u} \times \vec{v} = \left[\hat{k} - \left(\frac{\partial f}{\partial y} \right) \hat{j} - \left(\frac{\partial f}{\partial x} \right) \hat{i} \right] u_x v_y$$

By normalizing $\vec{N}$ and recognizing that the scalar magnitudes u_x, v_y cancel out, we arrive at the unit normal vector independent of the coordinate increments:

$$\hat{n}(x, y, z) = \frac{-\left(\frac{\partial f}{\partial x} \right) \hat{i} - \left(\frac{\partial f}{\partial y} \right) \hat{j} + \hat{k}}{\sqrt{\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 + 1}}$$

This vector is the unit normal to the surface $z = f(x, y)$ at any point (x, y, z) on the surface, which is required to evaluate $\vec{E} \cdot \hat{n}$ in Gauss's Law calculations.